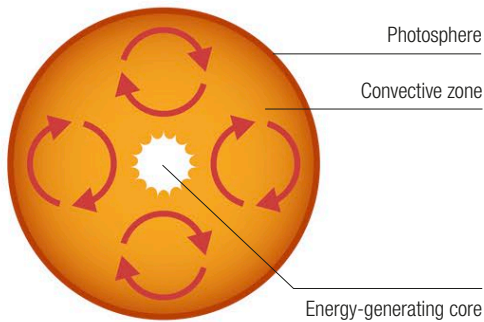


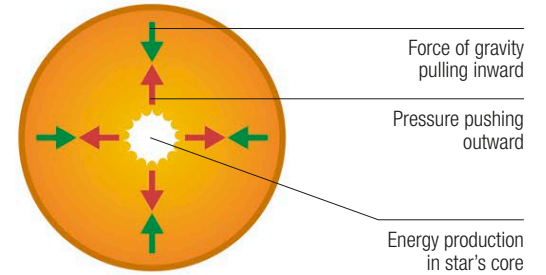


△ **Inside a red dwarf**

Inside a low-mass star (a red dwarf), the star's interior is mostly too dense for photons to penetrate far without being reabsorbed. Consequently energy is instead carried all the way to the surface by convection cells.

Forces inside stars

Whatever the mass of a star, two opposing forces keep it in existence. These are gravity, acting inward, and a pressure force, acting outward. Normally the opposing forces inside a star are in equilibrium, so it maintains its size over long periods of time. But if something causes the forces to become imbalanced, the star will change size. For example, the cores of most stars heat up toward the ends of their lives: the extra heat boosts the outward pressure, so the star swells into a giant or supergiant star.



△ **A star in equilibrium**

During most of the life of most stars, the inward-pulling force of gravity is exactly balanced by the outward-acting pressure, and the star maintains its size. If the forces get out of balance, the star is destined to either shrink or swell.

